

Multi-Step Deductive Reasoning Over Natural Language: An Empirical Study on Out-of-Distribution Generalisation

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Abstract

Combining deep learning with symbolic logic reasoning aims to capitalize on the success of both fields and is drawing increasing attention. Inspired by DeepLogic, an end-to-end model trained to perform inference on logic programs, we introduce IMA-GloVe-GA, an iterative neural inference network for multi-step reasoning expressed in natural language. In our model, reasoning is performed using an iterative memory neural network based on RNN with a gated attention mechanism. We evaluate IMA-GloVe-GA on three datasets: PARARULES, CONCEPTRULES V1 and CONCEPTRULES V2. Experimental results show DeepLogic with gated attention can achieve higher test accuracy than DeepLogic and other RNN baseline models. Our model achieves better out-of-distribution generalisation than RoBERTa-Large when the rules have been shuffled. Furthermore, to address the issue of unbalanced distribution of reasoning depths in the current multi-step reasoning datasets, we develop PARARULE-Plus, a large dataset with more examples that require deeper reasoning steps. Experimental results show that the addition of PARARULE-Plus can increase the model's performance on examples requiring deeper reasoning depths. The source code and data are available at <https://github.com/Strong-AI-Lab/Multi-Step-Deductive-Reasoning-Over-Natural-Language>.

Keywords

Gated attention, Out-of-distribution generalisation, Deductive reasoning, Multi-step reasoning

1. Introduction

Symbolic reasoning and deep learning remain two cornerstones in AI with profound yet divergent consequences. Indeed, symbolic approaches, equipped with various logic languages for knowledge representation and inference, have been the dominant paradigm in problem solving and reasoning. Deep learning approaches, through superior ability to capture rich semantic features from complex signals, triumph in tasks that usually require more intuitive and automatic judgements. A growing interest in AI amounts to harnessing the power from both schools, while mitigating each other's weaknesses. First, symbolic reasoning were suitable only when the task at hand, along with all contextual knowledge, can be encoded by rigorous and

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Table 1

Logic programs written in Prolog with different reasoning depths. These examples were sampled from [1].

1 Step	2 Steps	3 Steps
$g(L, S) :- x(S, L).$	$x(G, B) :- k(G, B).$	$p(P, R) :- b(R, P).$
$x(a, m).$	$k(Z, V) :- g(Z, V).$	$b(A, L) :- a(A, L).$
$y(X) :- r(X).$	$g(k, k).$	$a(W, F) :- v(F, W).$
$p(h).$	$e(k, s).$	$v(t, i).$
$s(t, v).$	$p(L, G) :- v(G, L).$	$l(D) :- t(D).$
$? g(m, a). 1$	$? x(k, k). 1$	$? p(t, i). 1$
$? g(a, m). 0$	$? x(k, s). 0$	$? p(i, t). 0$

structured logic expressions, which is itself a formidable obstacle. Then, deep learning relies on neural networks which have not demonstrated the ability to perform iterative, multi-step reasoning, which has a gap in making them suitable tools for reasoning. Towards a trainable reasoner that is able to perform complex real-world reasoning tasks, it is important to (1) facilitate end-to-end reasoning by enabling multi-step reasoning and (2) bypass logic-based knowledge representation and make inferences directly from natural language inputs. The goal of this paper is to explore possibilities around these two objectives. We now present the research background in detail.

Logic programs: Reasoning with logic programs is one of the key questions in AI. Here a knowledge base consists of a number of *rules*, i.e., (universally quantified) implications where the antecedent is a conjunction of literals and the consequent is an atom, and observed *facts*, i.e., ground atoms. The task would specify a *question* which is another ground atom and asks if the question logically follows from the knowledge base. Table 1 illustrates several archetypal examples of reasoning tasks in logic programs (as in [1]). The rules, facts, and questions are expressed in predicate logic where variables are capitalised (such as L, S) and constants are in small case (such as a, m). The three columns show logic programs of different reasoning *depths*. For example, the first column contains rule “ $g(L, S) :- x(S, L)$ ”, fact “ $x(a, m)$ ”, and two questions (starting with ?) at the bottom rows. Semantically, the rule expresses that $g(L, S)$ holds whenever $x(S, L)$ holds for any constants L and S . From $x(a, m)$, a simple unification followed by a 1-step forward chaining inference derives $g(m, a)$, which answers the first question positively [2]. On the other hand, the second column contains rules “ $x(G, B) :- k(G, B)$ ” “ $k(Z, V) :- g(Z, V)$ ”, and fact “ $g(k, k)$ ”. It takes two forward chaining steps to establish “ $x(k, k)$ ” and thus has depth 2.

DeepLogic: The ability to conduct iterated inference for multi-step reasoning tasks such as the ones above is viewed as an unchallenged strength of rule-based inference algorithms. Yet recent advancements in deep learning techniques have challenged this view. DeepLogic, introduced in [1], is an RNN-based neural network for solving reasoning tasks of logic programs. The model encodes logic programs at the character level and is trained on 12 different types of logic programs, without explicitly applying any symbolic inference algorithm. Through a series of experiments, DeepLogic has demonstrated abilities to handle tasks that require multi-step reasoning (up to a certain small depth).

Reasoning in natural language: The abilities demonstrated by DeepLogic has given hope

for similar neural networks to perform more general reasoning tasks. In particular, neural network’s key strengths involve the ability to extract rich syntactical and semantic feature from free-flowing texts, expressed in natural language. Indeed, PARARULES, introduced in [3], is a multi-step reasoning dataset expressed in natural language¹. Each sample in PARARULES resembles a logic program in the style of Table 1, except that the knowledge base (rules and facts) and questions are expressed in natural language (See Figure 2). We thus aim to explore end-to-end neural-based multi-step reasoners over natural language using PARARULES as a testing platform, while addressing three issues:

(1) Existing models, including DeepLogic and other RNN-based baseline models, have room for improvement in terms of their reasoning abilities over natural language. The vanilla GRU/LSTM model might not handle well the multi-step reasoning tasks over logic programs from [1] and natural language from Table 4. DeepLogic shows that with the help of GRU and dot-product attention, the model can learn to reason over logic programs from [1] and natural language from Table 4. However, DeepLogic does not show the best performance on Table 4. Dynamic memory network with gated attention has shown remarkable performance on the bAbI deductive reasoning task [4]. Our first contribution is to introduce IMA-GloVe-GA, an iterative neural inference network that combined DeepLogic with gated attention, for multi-step reasoning tasks. Our model achieves the best test accuracy among the RNN-based models on the PARARULES dataset. The test accuracy of our model is on average 7.8 percentage points higher than that of DeepLogic (from Table 4. IMA-GloVe-GA is our model, and IMA-GloVe is from DeepLogic).

(2) *Out-of-distribution (OOD) generalisation* means the test set has a distribution that is unknown or different from the distribution of the training set. In multi-step reasoning tasks, OOD generalisation means that (1) the model is able to reason for cases that have a depth greater than the depths of the instances it was trained on, and (2) the model is able to handle samples with shuffled rules from the training instances. Shuffling here means permuting the rules in the knowledge base. Being able to handle OOD is a crucial indicator of a model’s reasoning capabilities. In [3], pretrained RoBERTa-Large [5] achieves good performance on the PARARULES dataset (See Table 4). However, it is unclear whether the model indeed performs multi-step reasoning to the extent that it handles OOD test examples. Through a series of experiments, we show that RoBERTa-Large overfits and fails to generalise on examples with shuffled rules. This demonstrates that RoBERTa-Large over-utilises the ordering of rules. On the other hand, our IMA-GloVe-GA outperforms RoBERTa-Large and DeepLogic, when the models are trained on a dataset with fewer examples and unshuffled rules and are tested on a larger dataset with more relations and entities and shuffled rules (See Table 5).

(3) CONCEPTRULES V1 and V2 [6, 7] are natural-language-based multi-step reasoning datasets. Similar to PARARULES, the CONCEPTRULES datasets also contain samples that require deep reasoning steps (depth up to 3), and thus are suitable alternatives when evaluating models’ abilities for multi-step reasoning. A common issue with all three existing datasets (PARARULES, CONCEPTRULES V1 & V2), however, lies in their unbalanced distributions over reasoning depths. They have much fewer examples that require deep reasoning (depth ≥ 2) than examples that require shallow reasoning (See Table 2). To address the issue of depth

¹<https://allenai.org/data/rulemaker>